

Yuanbo Li

Brown University

PERSONAL INFORMATION

Yuanbo Li
Brown University
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Homepage: <https://liyb2002.github.io/>

GitHub: <https://github.com/Liyb2002>

Google Scholar: <https://scholar.google.com/citations?user=s1CcsUAAAAJhl=en>

RESEARCH INTERESTS

Computer Graphics, 3D Modeling, Computer-Aided-Design, Vision-Language Models, Neural-Symbolic Reasoning

EDUCATION

Ph.D. in Computer Science, Brown University 2024–Present
Advisor: Daniel Ritchie

M.S. in Computer Science, Brown University 2022–2024

B.A. in Mathematics, Columbia University 2018–2022

PUBLICATIONS

Journal Papers

[J1] **Li, Y.**, Manfredi, G*, Kriel, H*, Hao, C., Xu, X., Bousseau, A., Ritchie, D.
CADrawer: Autoregressive CAD Generation from 3D Sketches. Computer Graphics Forum (Proc. Eurographics), 2026.

[J2] **Li, Y.**, Ma, T., Aljumayaat, Z., Ritchie, D.
PossibleImpossibles: Exploratory Procedural Design of Impossible Structures. Computer Graphics Forum (Proc. Eurographics), 2024.

Conference and Short Papers

[C1] Kwon, H., **Li, Y.**, Ye, X., Muna-McQuay, P., Yin, L., Tompkin, J.
Active Appearance and Spatial Variation Can Improve Visibility in Area Labels for Augmented Reality. IEEE VIS, 2024.

Preprints and Submissions

PLLM: Pseudo Label Methods for LLM CAD Generation.

Li, Y., Shu, D., Chen, Y., Klenk, M., Ritchie, D.

Under submission.

Learning Object Placement Programs for Indoor Scene Synthesis with Iterative Self-Training.

Chang, A., Wang, K., **Li, Y.**, Savva, M., Chang, A. X., Ritchie, D.

Arxiv

PROFESSIONAL EXPERIENCE

Toyota Research Institute

Summer 2025

Research Intern

- Conducted research on LLM-based CAD program generation
- Designed and implemented a self-supervised pipeline for synthetic CAD data generation and LLM fine-tuning
- Presented LLM and CAD related research progress to the TRI research team

ARPA Technology

Summer 2022

Software Engineer Intern

- Designed and implemented behavior-driven testing pipelines for a distributed blockchain RNG system
- Assisted in debugging and validating on-chain contract logic and deployment workflows

Tencent

Summer 2021

Technical Product Manager Intern

- Contributed to architecture design for blockchain-based IPFS infrastructure
- Authored 10+ pages of technical documentation and a 30+ page whitepaper

ARPA Technology

Summer 2020

Software Engineer Intern

- Developed smart contracts for DeFi protocols using OpenZeppelin libraries
- Implemented behavior-driven tests for a onchain DeFi protocol

Tencent

Summer 2019

Product Manager Intern

- Produced technical documentation and requirement reports for WeChat platform features. Assisted in platform management.

PATENTS

[P1] **Pseudo-Label Learning Method for Training LLMs to Generate CAD Program Sequences**

Yuanbo Li, Dule Shu, Yanxia Zhang, Yanying Chen, Matt Klenk

U.S. Provisional Patent Application No. 63/863,263, Aug 2025.

STUDENT MENTORING

Xiaoxi Yang Research Assistant, Brown University

Chengye Hao M.S. in Computer Science, Brown University (expected 2026)

Bilal Tariq Undergraduate Student, Amherst College (expected 2027)

TEACHING

Teaching Assistant

Advanced Graphics, Brown University, Spring 2024

Teaching Assistant

Calculus III, Columbia University, Fall 2021

SELECTED TALKS

CAD Generation using LLMs, Toyota Research Institute, 2025
Sketch to CAD, Brown Visual Computing, 2024
Generating 3D Impossible Structures, Eurographics, 2024
Inverse Procedural Modeling, Brown Visual Computing, 2023

SERVICES

Reviewer, IEEE TVCG (2024)